THE YEN CARRY TRADE

Market Analysis & Risk Assessment

OMER GOLDSTEIN

Data Analyst | Financial Markets

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WHAT IS THE YEN CARRY TRADE?

Borrow low, invest high — profit from the interest rate differential

STEP 1

Borrow in Japan

¥15,000

≈ $100 USD

1% interest/yr

→

Convert JPY → USD

STEP 2

Invest in USA

$100

e.g. US Treasuries

5% return/yr

YOU PAY (JPY LOAN)		YOU EARN (USD INVEST)	NET PROFIT
−$1.00	+	+$5.00	$4.00
1% on $100		5% on $100	4% carry spread

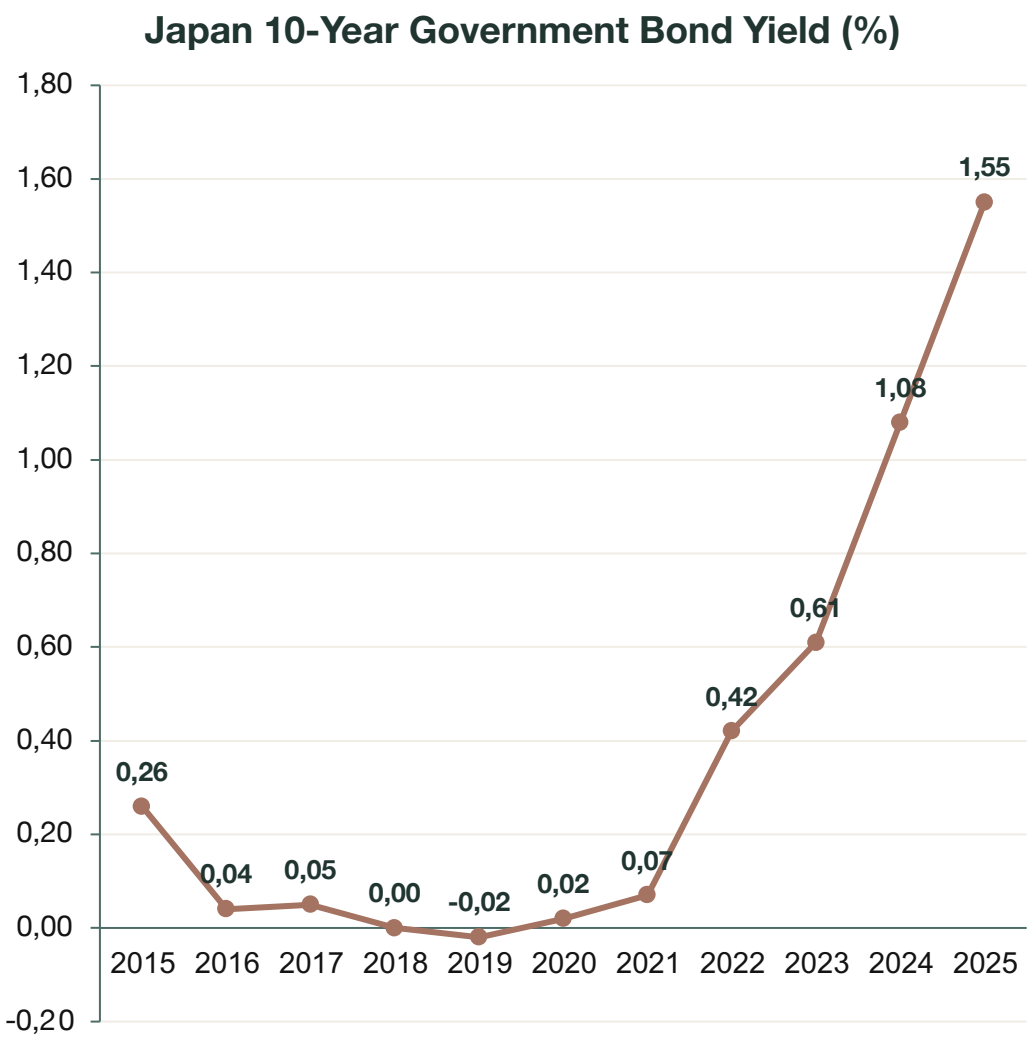
⚠ Risk: if the yen strengthens sharply, your $100 buys fewer yen when you repay — wiping out the profit.

THE END OF ZERO INTEREST RATES

Is the Yen Carry Trade over?

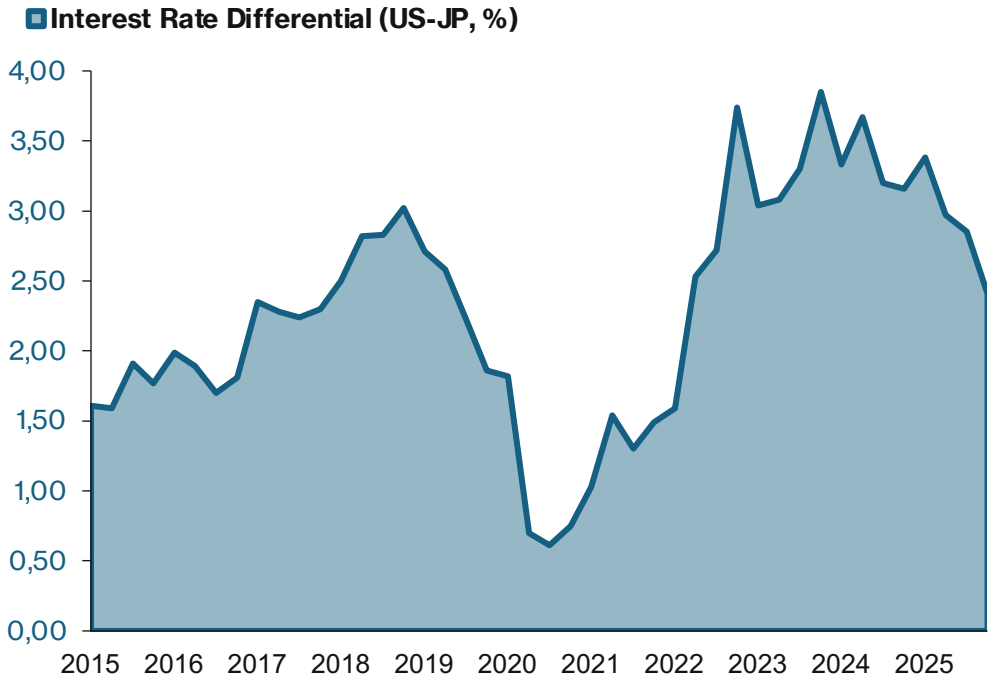
Japan's benchmark rate rose sharply after a decade near zero.

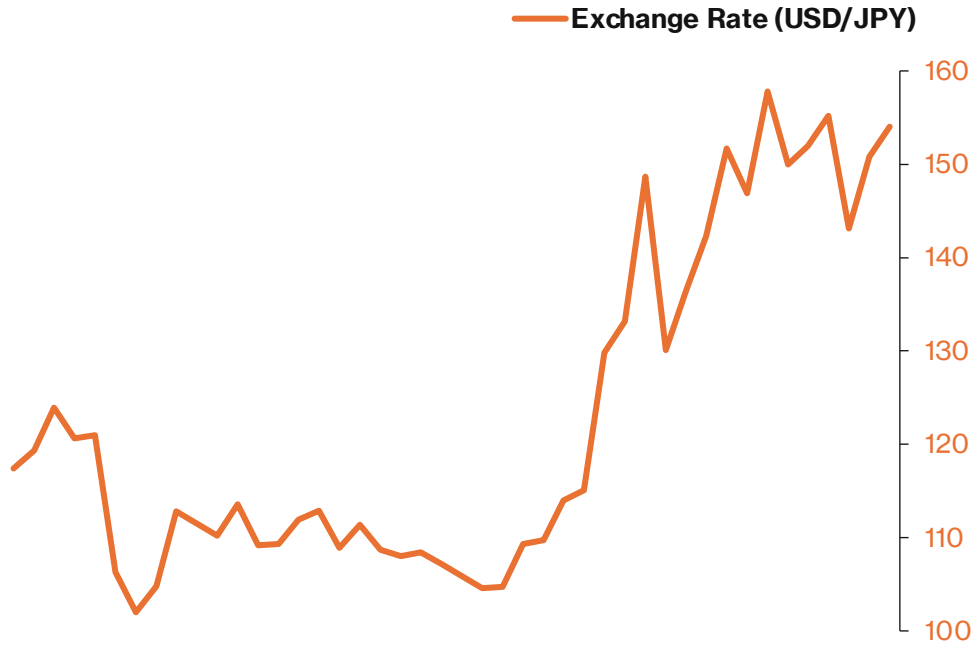
- Key shift: 2023–2025
- First rate hikes since 2007
- Direct impact on carry trade viability



RATE DIFFERENTIAL & USD/JPY

How the rate spread drives the currency





INTEREST RATE DIFFERENTIAL

US–Japan spread averaged ~2% over the period, keeping the carry trade profitable.

USD/JPY EXCHANGE RATE

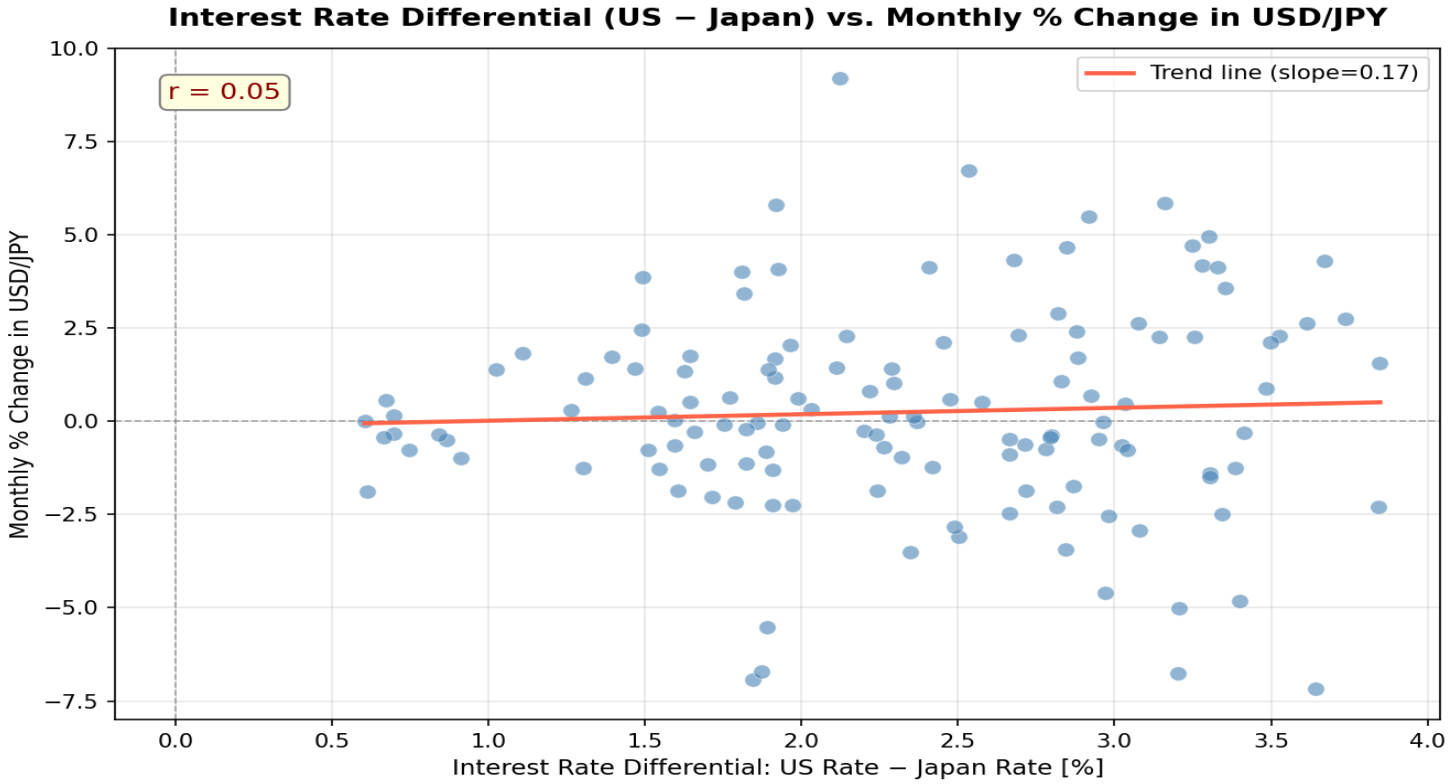
Persistent Yen depreciation throughout the period — a tailwind for carry traders.

SHORT-RUN DISCONNECT

Rates don't drive USD/JPY in the short run

KEY FINDING

Despite rising rates in Japan, short-run USD/JPY movements show almost no linear relationship with the rate differential (correlation = 0.05).



Rate differential vs. USD/JPY — short-run correlation

LONG-RUN RELATIONSHIP

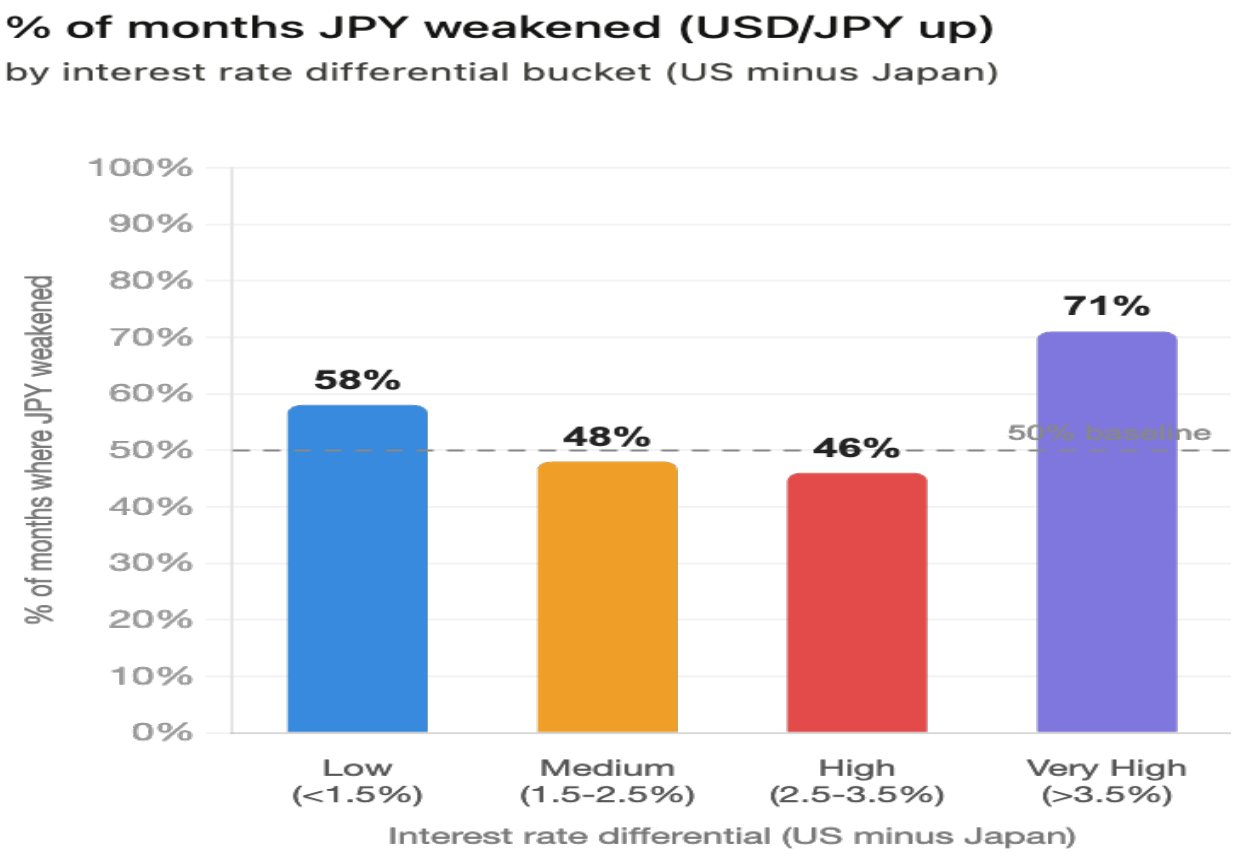
Over longer horizons, rates matter

KEY FINDING

When the interest rate differential widens, the probability of yen depreciation rises sharply — especially at the "Very High" spread level.

INSIGHT

Unlike the short run, long-term FX moves align with rate fundamentals — supporting the carry trade logic.



THE REAL DRIVER OF CARRY PROFITS

It's the currency, not the rate

NUMERICAL EXAMPLE

$100 → $169

Carry trade return, 2015 → 2025

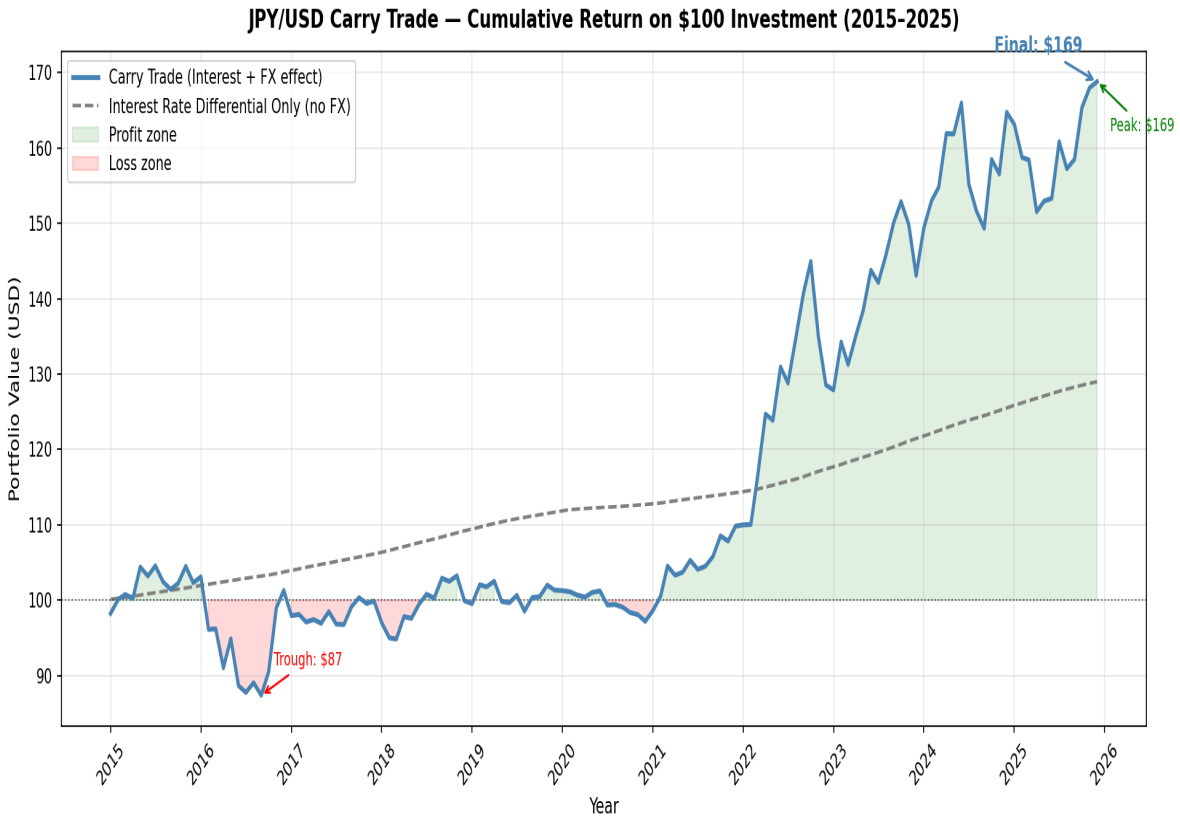
WHERE THE PROFIT CAME FROM

+$29 Interest rate differential (US–JP spread)

+$40 **FX effect — yen depreciation (2022–2024)**

TAKEAWAY

The carry trade is meaningfully profitable only when accompanied by a strong FX move — not from the rate differential alone.



Cumulative return: interest only vs. interest + FX

IS THE YEN CARRY TRADE OVER?

Three conclusions from the data

01 TAIL RISK

The Yen Is Not "Stable"

On Jul 31, 2024, BOJ hiked to just 0.25%.
Within one week:

USD/JPY: 160 → 142	**–11%**
Nikkei (Aug 5, worst since 1987)	**–12.4%**
Toyota profit QoQ	**–20%**
VIX spiked above	**65**

The yen is the most explosive carry currency in the world. Tail risk is not theoretical.

02 RISK MANAGEMENT

How to Survive the Shocks

Low leverage

If yen appreciates 10% against you:

3x leverage	**–30%**
5x leverage	**–50%**
10x leverage	**–100%**

Diversified funding

Mix JPY with CHF and EUR.

Duration matching

Short yen loans ≠ long illiquid assets.

The survivors of Aug 2024 weren't the smartest — they were the least leveraged.

03 OUTLOOK

Short-Term vs. Long-Term

Short term (6–12 months)

Weak yen continues. BOJ 0.5% vs Fed ~4.5% (~4% spread) keeps carry alive. Yen weakens in 71% of months when spread > 3.5%.

Long term (3–5 years)

Easy-money era is over. Rate gap stays positive but narrower. Buy-and-hold now requires active management, hedging, and timing.

The carry trade dies slowly.

Sources: FRED · Bank of Japan · Investing.com · Analysis: Jan 2015 – Dec 2025

APPENDIX

Data sources, methodology, and dataset

DATA SOURCES

- FRED — Federal Reserve Economic Data (interest rates)
- Investing.com — USD/JPY exchange rate

DATA RANGE

- January 2015 – December 2025
- Monthly frequency

DATASET

Full dataset and analysis available on GitHub:
github.com/omergoldstein24-debug/yen-carry-trade-analysis

METHODOLOGY

- Constructed a unified dataset from multiple raw sources
- Calculated interest rate differential (US – Japan)
- Computed monthly percentage change in USD/JPY
- Grouped observations into buckets by rate differential
- Measured correlation using Pearson coefficient
- Data cleaning and transformation in Python
- Visualization with matplotlib and seaborn